NISR AgriShield Rwanda: Raw Data & Provenance Report

Official Technical Dossier of Datasets, Spatial Sources, Direct Links & Variables Fused for SSIT Solar Pump Prioritization

Executive Summary & Mandate: AgriShield Rwanda is a spatial decision-support engine developed for the **National Institute of Statistics of Rwanda (NISR) Hackathon 2026** to support the **Ministry of Agriculture and Animal Resources (MINAGRI)** and the **Rwanda Agriculture and Animal Resources Development Board (RAB)**. The system targets Small-Scale Irrigation Technology (SSIT) 50% solar pump government subsidies by harmonizing **four authoritative data pillars**: (1) Official Rwandan agricultural microdata (NISR SAS), (2) Socio-economic household poverty indices (NISR EICV), (3) Earth Observation & Climate Reanalysis (ECMWF ERA5-Land & CHIRPS), and (4) Administrative boundary cartography (GADM 4.1). This document provides the full provenance, definitions, and direct access links for all raw data utilized across Rwanda's 30 districts and 422 sectors.

1. Administrative Geographic Boundaries (GADM 4.1)

High-resolution spatial multi-polygon vectors for Rwanda were extracted from the **Global Administrative Areas (GADM)** database (Version 4.1), the standard global administrative boundary reference maintained by UC Davis. These vector layers define the territorial geometries used to compute exact geometric centroids, raster zonal statistics, and client-side 3D extruded visualizations.

Administrative Level	Entities Count	Local Unit	Coordinate System	Direct Download & Source URL
Level 2 (Districts)	30 Districts	Akarere	WGS 84 (EPSG:4326)	geodata.ucdavis.edu/.../gadm41_RWA_2.json
Level 3 (Sectors)	416–422 Sectors	Umurenge	WGS 84 (EPSG:4326)	geodata.ucdavis.edu/.../gadm41_RWA_3.json

2. Agricultural Exposure & Cropland Baselines (NISR SAS)

Agricultural cultivation areas and irrigation penetration rates are derived from the **Seasonal Agricultural Survey (SAS)** conducted annually across Agricultural Seasons A, B, and C by the **National Institute of Statistics of Rwanda (NISR)** in collaboration with **MINAGRI**. SAS microdata captures crop hectarage, farming typologies, and water source availability.

Variable	Units	National Aggregate	Technical Definition & Role	Official Data Portal Link
total_cropland_ha	Hectares (ha)	896,180 ha	Total physical crop cultivation area per district/sector.	NISR SAS Publications Portal
irrigated_pct	Percent (%)	5.6% (avg)	Existing formal or informal irrigation infrastructure coverage.	NISR Microdata Catalog: SAS
unirrigated_ha	Hectares (ha)	845,988 ha (94.4%)	Total rainfed land strictly exposed to rainfall failure: *cropland × (1 - irrigated_pct)*.	MINAGRI SSIT Strategic Plan

3. Socio-Economic Equity & Poverty Profiling (NISR EICV)

To ensure that subsidized pumps are not disproportionately captured by wealthier commercial farms, the optimization engine fuses headcount poverty rates from the **Integrated Household Living Conditions Survey (EICV)** series (EICV5 / EICV6) published by NISR. Under Rwanda's NST2 vision, targeting priority is awarded to climate-vulnerable zones with high poverty shares.

Metric Variable	Source Survey	Range Across Rwanda	Policy Role in AgriShield	Direct Portal Reference
eicv_poverty_pct	NISR EICV5 / EICV6 Poverty Profile	8.2% (Kicukiro) to 46.2% (Gisagara)	Equity weighting: ensures 50% capital subsidies reach low-income agrarian smallholders.	NISR EICV Poverty Series EICV Microdata Catalog

4. Climate Reanalysis & Earth Observation (ECMWF ERA5 & CHIRPS)

Climate and agronomic drought risks are driven by high-resolution atmospheric reanalysis and satellite precipitation data. AgriShield queries the **European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5-Land Reanalysis** (spatial resolution ~9 km) and **UCSB Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS)** via the Open-Meteo Historical Archive API. The analysis period covers the real **2026 Season A agricultural growing window** (January 1, 2026 – March 31, 2026).

Climate Feature	Units	Physical Sensor / Model	Rwanda Observed Range	Direct Access Endpoint & Link
real_precip_mm	Millimeters (mm)	CHIRPS Daily + ERA5-Land	254.7 mm (Kayonza) to 433.6 mm (Bugesera)	Open-Meteo ERA5 API Docs
real_temp_c	Celsius (°C)	ERA5 2m Air Temperature	18.9 °C to 21.2 °C (Seasonal mean)	Copernicus CDS: ERA5-Land
real_soil_moisture_m3_m3	m^3/m^3	ERA5-Land (0–7 cm topsoil)	0.150 (Ngoma drought) to 0.386 (Gasabo)	ECMWF ERA5 Overview
dry_spell_days	Days	Precipitation < 1.0 mm count	11 days (Musanze) to 42 days (Bugesera)	UCSB CHIRPS Database
empirical_moisture_stress	Index (0–1)	Physical Deficit Formula	0.255 (Low) to 0.805 (Extreme Stress)	*Supervised Target for ML Training*

Empirical Moisture Stress Index Formula:

$$\text{Moisture Stress} = 0.55 \times [\,(0.42 - \text{SoilMoisture}) / 0.25\,] + 0.45 \times [\,\text{DrySpellDays} / 60\,]$$ *(Bounded between 0.05 and 0.98)*

- **0.42 m^3/m^3:** Baseline root-zone soil saturation field capacity for Rwandan volcanic/clay-loam soils.
- **60 Days:** Maximum duration of critical vegetative flowering and grain-filling window during Season A.

5. Master Feature Dictionary & Model Input Matrix

Feature Name	Type	Level	Source Institution	Role in ML / Policy Optimization
NAME_2	Categorical	District	GADM / MINALOC	Administrative primary key for 30 districts
NAME_3	Categorical	Sector	GADM / MINALOC	Micro deployment key for 422 sectors
total_cropland_ha	Numeric (ha)	Both	NISR SAS	Agricultural scale and sector density weighting
irrigated_pct	Percent (%)	Both	NISR SAS / RAB	Infrastructure baseline to isolate rainfed deficit
unirrigated_ha	Numeric (ha)	Both	Calculated (SAS)	Cropland Exposure component (w2 in Utility formula)
eicv_poverty_pct	Percent (%)	Both	NISR EICV	Socio-economic vulnerability component (w3 in Utility)
real_precip_mm	Float (mm)	Both	ECMWF / CHIRPS	ML feature predicting seasonal water deficit
real_temp_c	Float (°C)	Both	ECMWF ERA5	ML feature for evapotranspiration demand
real_soil_moisture_m3_m3	Float	Both	ECMWF ERA5-Land	Top primary predictor of plant water stress
dry_spell_days	Integer	Both	ECMWF / CHIRPS	Rainfall irregularity metric capturing acute shocks
predicted_stress_prob	Probability	Both	LightGBM / XGBoost	Climate Risk component (w1 in Utility formula)
ssit_priority_score	Score (0–1)	Both	AgriShield Engine	Final allocation ranking: *w1·Risk + w2·Crop + w3·Pov*

6. Repository Pipeline & Reproducibility Manifest

All data pipelines, raw inputs, and processed baseline tables are checked into the repository and fully reproducible. The following manifest links code pipelines with their inputs and outputs:

File / Script	Type	Description
src/download_boundaries.py	Ingestion Script	Downloads Level 2 (Districts) and Level 3 (Sectors) GeoJSON polygons directly from GADM 4.1.
src/fetch_real_climate_and_sas.py	Processing Script	Extracts geometric centroids, queries ECMWF ERA5-Land reanalysis via Open-Meteo, merges NISR SAS & EICV baselines.
src/model_pipeline.py	ML Engine	Trains LightGBM with Stratified 5-Fold CV, computes TreeSHAP feature importance, and outputs predictive risk probabilities.
data/raw/rwanda_districts_lvl2.json	Raw GeoJSON	Official boundary coordinates for Rwanda's 30 districts (122 KB).
data/raw/rwanda_sectors_lvl3.json	Raw GeoJSON	Official boundary coordinates for Rwanda's 422 sectors (593 KB).
data/processed/sas_district_baseline.csv	Processed Table	Merged master baseline table for 30 districts with 13 features (3.2 KB).
data/processed/sas_sector_baseline.csv	Processed Table	Merged master baseline table for 422 sectors with 13 features (29.5 KB).
data/processed/district_predictions.csv	Model Output	District predictions with SHAP values and predicted climate stress probabilities (3.5 KB).
data/processed/sector_predictions.csv	Model Output	Sector predictions with SHAP values and predicted climate stress probabilities (33.3 KB).

7. Institutional References & Direct Links Directory

1. **National Institute of Statistics of Rwanda (NISR):** https://www.statistics.gov.rw — Primary source for Seasonal Agricultural Survey (SAS) and EICV poverty series.
2. **NISR Microdata Archive Catalog:** https://microdata.statistics.gov.rw — Official catalog for household microdata datasets.
3. **Ministry of Agriculture and Animal Resources (MINAGRI):** https://www.minagri.gov.rw — SSIT policy guidelines and NST2 irrigation targets.
4. **Rwanda Agriculture and Animal Resources Development Board (RAB):** https://www.rab.gov.rw — Field implementation partner for SSIT subsidized pump delivery.
5. **European Centre for Medium-Range Weather Forecasts (ECMWF):** https://www.ecmwf.int & Copernicus Climate Change Service ERA5-Land Reanalysis.
6. **UCSB Climate Hazards Center (CHIRPS):** https://www.chc.ucsb.edu/data/chirps — Quasi-global high-resolution precipitation.
7. **GADM Administrative Boundaries:** https://gadm.org — Spatial administrative subdivisions for Rwanda (Level 0, 1, 2, 3).